HUMAN VERIFICATION DOSSIER

for Amber Classification of AI-Assisted Research

Linear Nonequilibrium Information Suffices for Fault-Tolerant Subquantum Search

Prepared for

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Purpose of this dossier

This workbook is designed to make the human validation required for an amber AI-use classification explicit, reproducible, and easy to audit. It does not guarantee that an editor will assign an amber classification; the journal retains that judgement.

Manuscript version reviewed: Foundations_of_Physics_submission_Stay_SN_revised_ready.zip (8 October 2026)

1. How to use this dossier

Springer Nature currently treats AI use by risk rather than by tool. Amber use includes AI contributions to analytical or interpretive work, provided the use is transparent, the outputs are independently verified, and scholarly judgement remains demonstrably human. The present manuscript is an amber candidate because ChatGPT contributed substantial reasoning and drafting. It risks red classification if the central analysis remains effectively delegated to the model or cannot be independently defended by the author.

> **The governing test**
>
> At the end of this process, you should be able to defend every equation, assumption, citation, novelty claim, interpretation, and conclusion without relying on ChatGPT as an authority.

Status notation

- PASS: you independently reproduced or checked the result and retained evidence.
- REVISION NEEDED: the idea is defensible, but the manuscript must be corrected or narrowed.
- FAIL: the result, citation, or claim cannot presently be defended.
- N/A: the item does not apply; record why.

Suggested evidence packet

[ ] A handwritten or independently typed derivation of every calculation in Sections 2-6 of the manuscript.

[ ] A citation audit showing the exact source location supporting each externally sourced claim.

[ ] A dated literature-search log documenting the novelty search.

[ ] A list of all manuscript changes made after independent verification.

[ ] A final signed authorship/accountability statement and updated AI disclosure.

[ ] A retained copy of the ChatGPT conversation or an equivalent interaction log.

Submission gate

> **Do not submit while a critical item is unresolved**
>
> A transparent declaration alone does not convert unverified AI-generated reasoning into acceptable amber use. The verification and human judgement must actually have occurred.

2. Springer Nature amber criteria mapped to this paper

Criterion	What it means here
AI contribution	Amber permits AI to contribute to analytical or interpretive processes. Here ChatGPT proposed core constructions and drafted substantial text, so the use is not merely green.
Transparency	The tool, model, timing, purposes, and substantive contributions must be described accurately.
Validation	The author must independently verify all mathematics, physics, complexity claims, references, and novelty.
Human judgement	The author must decide which assumptions are acceptable, what theorem is actually proved, what is novel, and what conclusions are warranted.

Criterion	What it means here
Reproducibility	Inputs, derivations, source locations, corrections, and validation steps should be documented well enough for independent assessment.
Accountability	The human author remains solely responsible; ChatGPT is not an author and cannot bear responsibility.
Red-line risk	The use becomes red if AI-generated hypotheses, analyses, or conclusions are presented as human-derived without adequate oversight, or if core reasoning remains unverifiable or effectively delegated.

Official policy pages

Springer Nature - AI use in research practice: https://www.springernature.com/gp/policies/editorial-policies/using-ai-in-research

Springer Nature - AI use in manuscript preparation: https://www.springernature.com/gp/policies/editorial-policies/ai-manuscript-preparation

Springer Nature - editor AI assessment guidance: https://www.springernature.com/gp/policies/editorial-policies/ai-editor-guidance

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____ _

3. Revision audit: issues addressed in the revised manuscript

The revised paper now addresses the earlier critical defects

This section records how each issue identified in Version 1.0 was repaired. The green status means "addressed in the manuscript," not "independently verified." Each mathematical and source claim must still pass the later modules before amber sign-off.

C1 - Rare-set lemma repaired

The revised lemma now defines r = ρ(B), assumes $0 < p < c \leq 1$ and $r \geq c$, applies data processing to obtain $D(\rho||\mu) \geq d(r||p)$, differentiates d with respect to r, and uses monotonicity only on $r > p$. The c = 1 endpoint is handled by continuity. Verify the proof line by line in Module 5.3.

C2 - Design and actual marked-state weights separated

The revised paper defines p* = 2^{-n} as the design value and p_f = w p* with w ∈ {0,1} as the actual marked-state weight. The same shell is prepared before the instance is known. Verify the no-witness and unique-witness cases separately in Modules 5.4, 5.8, and 5.9.

C3 - Full computation included in the fault bound

Assumption (A1) now defines L(n) = poly(n) to include state preparation, the reversible verifier, transfer into the readout code, syndrome operations, and bounded-strength segmentation of the pointer displacement. The proof uses $Pr_\mu(B_k) \leq L(n)q_k$ and therefore $Pr_\rho(\text{failure}) \leq 4L(n)q_k/p^*$. Verify Modules 5.17 and 5.18.

C4 - Error-transparent operator defined on the full physical Hilbert space

The revised paper specializes to the nondegenerate five-qubit perfect code. Its sixteen two-dimensional syndrome sectors exhaust the 32-dimensional physical Hilbert space. The general fixed-code variant is explicitly extended by zero on K⊥. Verify the code dimension, projector properties, and self-adjointness in Modules 5.11 and 5.12.

C5 - Transparent histories precisely delimited

The revised paper proves the one-fault identity, separately proves transparency for sequences that act only on the syndrome factor, defines those histories as transparent, and counts every unproved history as nontransparent. It no longer says that arbitrary products of individually correctable errors are harmless. Verify Modules 5.13 and 5.14.

C6 - Complexity consequence made explicitly randomized

The revised paper states the zero-versus-one promise, allows arbitrary solver behavior on multiple-witness derived instances, describes repeated randomized isolation and amplification, and separates SAT decision from witness recovery by standard self-reduction. Verify the original Valiant–Vazirani theorem and the self-reduction in Module 5.19.

C7 - Ideal unit-probability statement separated from noisy success

The revised paper uses the design shell A*, treats p_f = 0 and p_f = p* separately, states the packet-separation and one-dimensional guidance assumptions, and then bounds noisy failure only through the explicitly defined bad event B_k. Verify Modules 5.7–5.9 and the theorem proof.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

4. Manuscript claim map

Manuscript location	Claim	Verification
Section 2, resource definition	Fine-grained relative entropy is additive and invariant under exact reversible pilot-wave flow.	Modules 5.1–5.2
Lemma 1 and proof	If r ≥ c > p, rare-set selection costs at least d(c||p) = Ω(ln(1/p)).	Module 5.3
Section 3, design/actual weights	The verifier marks a pre-existing component; p_f is 0 or p* = 2^{-n}.	Module 5.4
Section 3, pointer dynamics	The von Neumann coupling translates only the marked component and gives the stated density and current.	Modules 5.5–5.6
Section 3, cumulative probability	C(Y(t),t) is conserved and labels the marked-state basin.	Modules 5.7–5.8
Section 4, shell and conditioning	The same design shell works on both promised cases and costs n ln 2 + ln 4.	Module 5.9

Manuscript location	Claim	Verification
Section 4, tail geometry	A smooth compactly supported packet can contain an exponential-tail region with O(1) shell width and O(n) location/displacement.	Module 5.10
Section 5, five-qubit code and Q-tilde	The syndrome sectors exhaust the physical Hilbert space and the replicated logical projector is a bounded self-adjoint projector.	Modules 5.11–5.12
Section 5, transparent histories	One-fault histories and syndrome-only sequences have exactly the ideal pointer translation; all others are charged to q_0.	Modules 5.13–5.14
Section 6, theorem proof	The full-circuit union bound and likelihood-ratio bound imply O(log n) concatenation levels and polynomial overhead.	Modules 5.15–5.18
Section 7	The complexity consequence is a randomized SAT decision consequence; search follows by self-reduction.	Module 5.19
Section 9	AI use, validation, and sole human accountability are disclosed accurately.	Sections 9–10

5. Detailed mathematical and physical verification

For each module, first reproduce the calculation yourself without looking at the completed derivation. Then compare line by line, consult the primary source where indicated, and sign the module. The derivations below are aids to checking; they are not substitutes for your independent judgement.

5.1. Equivariance and conservation of the density ratio

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Claim to verify: if ρ and $\mu = |\psi|^2$ obey continuity equations with the same Bohmian velocity field v, then $f = \rho/\mu$ is constant along trajectories.

$$\partial_t \rho + \nabla\cdot(\rho v) = 0, \qquad \partial_t \mu + \nabla\cdot(\mu v) = 0$$

Step	Statement / calculation	Reason or check
1	Set $f = \rho/\mu$ wherever $\mu > 0$.	Absolute continuity $\rho \ll \mu$ is needed for a finite relative entropy and a well-defined ratio.
2	Differentiate $\rho = f\mu$: $\partial_t\rho = \mu\partial_t f + f\partial_t\mu$.	Product rule.
3	Expand $\nabla\cdot(\rho v) = \nabla\cdot(f\mu v) = \mu v\cdot\nabla f + f\nabla\cdot(\mu v)$.	Product rule for divergence.
4	Insert the two expansions into $\partial_t\rho + \nabla\cdot(\rho v) = 0$.	Use the continuity equation for ρ.
5	Obtain $\mu(\partial_t f + v\cdot\nabla f) + f[\partial_t\mu + \nabla\cdot(\mu v)] = 0$.	Collect the terms proportional to μ and f.
6	The bracket vanishes by the continuity equation for μ.	This is the key “same velocity field” assumption.

Step	Statement / calculation	Reason or check
7	Where μ > 0, divide by μ: $\partial_t f + v\cdot\nabla f = 0$.	This is df/dt = 0 along a Bohmian trajectory.
8	Conclude f(q(t),t) = f(q(0),0).	Solve the first-order advection equation along characteristics.

[ ] Re-derive the result without copying the table.

[ ] Verify Valentini 2024, Eq. (21), and the surrounding discussion.

[ ] State how nodes μ = 0 are treated and confirm that the chosen pointer support avoids problematic nodes.

[ ] Confirm that the statement applies to the full joint configuration, including pointer, computer, syndrome apparatus, and environment.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.2. Invariance of the fine-grained relative entropy

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Claim to verify: $R(\rho||\mu) = \int \rho \ln(\rho/\mu)\, dq$ is constant under exact reversible pilot-wave evolution, subject to appropriate boundary conditions.

Step	Statement / calculation	Reason or check
1	Write $R(t) = \int \rho(q,t) \ln f(q,t)\, dq$.	Use $f = \rho/\mu$.
2	Differentiate under the integral: $dR/dt = \int[(\partial_t\rho)\ln f + \rho\partial_t \ln f]dq$.	Requires sufficient regularity and integrability.
3	Use $\partial_t\rho = -\nabla\cdot(\rho v)$.	Continuity equation.
4	Integrate the first term by parts: $\int -\nabla\cdot(\rho v)\ln f\, dq = \int \rho v\cdot\nabla \ln f\, dq$.	Boundary flux must vanish, or configuration space must have suitable periodic/closed boundaries.
5	Use $\partial_t \ln f = -v\cdot\nabla \ln f$.	This follows from df/dt = 0.
6	The two terms cancel exactly.	The remaining integrand is $\rho[v\cdot\nabla\ln f - v\cdot\nabla\ln f]$.
7	Conclude dR/dt = 0.	This is fine-grained invariance, not the coarse-grained H-theorem.

[ ] Check normalization of ρ and μ.

[ ] Check the boundary term explicitly for the pointer packet used later.

[ ] Distinguish fine-grained invariance from coarse-grained relaxation.

[ ] Decide whether “resource” should be measured in nats R or bits R/ln 2 and use one convention consistently.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL

Evidence / initials / date	_____

5.3. Rare-set lower bound: revised proof

Critical verification gate

Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Revised manuscript status

The missing hypothesis has been repaired. The verification task is now to reproduce the binary coarse-graining, derivative, monotonicity domain, endpoint convention, and asymptotic expansion without relying on the manuscript text.

Let $p = \mu(B)$ and $r = \rho(B)$. Coarse-grain the configuration to the binary outcome “inside B” versus “outside B.”

$$d(r||p) = r\ln(r/p) + (1-r)\ln[(1-r)/(1-p)]$$

Step	Statement / calculation	Reason or check
1	Let $p = \mu(B)$, $r = \rho(B)$, and assume $0 < p < c \le 1$ with $r \ge c$.	These are the exact hypotheses of the revised lemma.
2	Coarse-grain with $T(q) = 1_B(q)$. The pushforwards are Bernoulli(r) and Bernoulli(p).	Membership in B is a deterministic measurable map.
3	Data processing gives $D(\rho||\mu) \ge d(r||p)$.	Relative entropy cannot increase under a deterministic coarse-graining.
4	$d(r||p) = r\ln(r/p) + (1-r)\ln[(1-r)/(1-p)]$.	Binary relative entropy.
5	Differentiate: $\partial_r\, d(r||p) = \ln[r(1-p)/(p(1-r))]$.	Differentiate both terms and cancel the ±1 contributions.
6	For $r > p$, $r(1-p) > p(1-r)$, so $\partial_r\, d(r||p) > 0$.	The inequality simplifies exactly to $r > p$.
7	Because $r \ge c > p$, $d(r||p) \ge d(c||p)$.	Monotonicity is used only to the right of the minimum $r = p$.
8	For $c = 1$, interpret $(1-c)\ln[(1-c)/(1-p)]$ as 0 by continuity.	The limit $x\ln x \to 0$ as $x \to 0+$.
9	For fixed $c > 0$, $d(c||p) = c\ln(1/p) + O(1)$ as $p \to 0$.	The remaining c-dependent terms are bounded and $\ln(1-p) = O(p)$.
10	Therefore $D(\rho||\mu) = \Omega(\ln(1/p))$; for $r = c = 1$ the lower bound is exactly $\ln(1/p)$.	This is the lower bound used for deterministic marked-state selection.

Revised lemma to compare against your independent derivation

> **Revised statement**
>
> Let μ(B)=p and ρ(B)=r. If 0<p<c≤1 and r≥c, then D(ρ||μ)≥d(c||p), with the endpoint interpreted by continuity. For fixed c>0 and p→0 this is Ω(ln(1/p)).

[ ] Reproduce the data-processing step using the binary indicator 1_B.

[ ] Differentiate d(r||p) yourself and verify the sign only for r > p.

[ ] Check the c = 1 endpoint and the convention 0 ln 0 = 0.

[ ] Confirm that the application uses p = p* = 2^{-n} and a fixed target success c > p* for sufficiently large n; verify data processing from a trusted source.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____ _

5.4. Marked-state weight produced by the reversible verifier

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Claim to verify: the uniform superposition already contains every candidate component. Define p* = 2^{-n}, let w ∈ {0,1} be the witness count, and set p_f = w p*. The verifier marks, rather than creates, the unique witness component.

Step	Statement / calculation	Reason or check
1	Start with $\|+\rangle^{\otimes n}\|0\rangle = 2^{-n/2} \Sigma_x \|x\rangle\|0\rangle$.	Every candidate basis component is already present.
2	Apply U_f: $\|x\rangle\|0\rangle \mapsto \|x\rangle\|f(x)\rangle$.	The reversible verifier correlates the flag with the predicate.
3	The state is $\|\Psi_f\rangle = 2^{-n/2} \Sigma_x \|x\rangle\|f(x)\rangle$.	Linearity.
4	Define p* = 2^{-n}, w = Σ_x f(x) ∈ {0,1}, and p_f = w p*.	Separates the design value from the instance-dependent weight.
5	With $Q = I \otimes \|1\rangle\langle 1\|$, $\langle\Psi_f\|Q\|\Psi_f\rangle = 2^{-n}\Sigma_x$ f(x) = p_f.	Orthogonality of the computational basis removes cross terms.
6	No witness: w = 0 and p_f = 0.	There is no marked-state packet.
7	Unique witness: w = 1 and p_f = p*.	The marked component has the design weight.

[ ] Verify the reversible implementation of the chosen verifier has polynomial size.

[ ] Use “marked state” or “marked subspace” consistently; do not say the verifier creates a previously absent basis component.

[ ] Confirm that the notation p* and p_f is used consistently in the abstract, Sections 1–4, and the theorem proof.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.5. Unitary translation generated by H_M = a(t)Q p_y

> **Critical verification gate**
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Assume $\hbar = 1$, $p_y = -i\partial_y$, and that Q commutes with p_y. The interaction dominates other Hamiltonian terms during the measurement interval.

Step	Statement / calculation	Reason or check
1	Because $H(t) = a(t)Qp_y$ differs at different times only by a scalar, $[H(t),H(s)] = 0$.	No time ordering complication remains.
2	Define $G(t) = \int_0^t a(s)ds$.	Integrated coupling strength.
3	The unitary is $U(t) = \exp[-iG(t)Qp_y]$.	Integrate the Schrödinger equation.
4	Use $Q^2 = Q$ to decompose $U = (I-Q)\otimes I + Q\otimes\exp[-iGp_y]$.	Functional calculus for a projector.
5	For $p_y = -i\partial_y$, $\exp[-iGp_y]g(y) = \exp[-G\partial_y]g(y) = g(y-G)$.	The exponential of the derivative is a translation operator.
6	Therefore only the $Q = 1$ component shifts: $\sqrt{(1-p)}\|\Phi_0\rangle g(y) + e^{i\varphi}\sqrt{p}\|\Phi_1\rangle g(y-G)$.	Apply the decomposition to the two orthogonal system sectors.
7	If $G > 0$, the marked packet moves to the right. If $G < 0$, the later basin inequality reverses.	The sign convention must be checked explicitly.

[ ] Verify Valentini 2024, Eqs. (18)-(19), and Valentini 2002, Section 4.

[ ] Check the translation sign directly on a test function, such as $g(y) = \exp(iky)$.

[ ] State whether other Hamiltonian terms are neglected exactly or approximately.

[ ] Confirm Q is a projector rather than a general observable in the manuscript construction.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.6. Pointer density, current, and Bohmian velocity

> **Critical verification gate**
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

This calculation should be checked directly because the Hamiltonian is first order in momentum, not the usual $p^2/2m$ kinetic Hamiltonian.

Step	Statement / calculation	Reason or check
1	The Schrödinger equation is $i\partial_t\Psi = aQ(-i\partial_y)\Psi$, hence $\partial_t\Psi = -aQ\partial_y\Psi$.	Cancel i and use $p_y = -i\partial_y$.
2	Take the Hermitian conjugate: $\partial_t\Psi^\dagger = -a(\partial_y\Psi^\dagger)Q$.	Q is Hermitian and a is real.
3	Let $P = \Psi^\dagger\Psi$. Then $\partial_t P = (\partial_t\Psi^\dagger)\Psi + \Psi^\dagger\partial_t\Psi$.	Product rule.
4	Substitute the evolution equations: $\partial_t P = -a[(\partial_y\Psi^\dagger)Q\Psi + \Psi^\dagger Q\partial_y\Psi]$.	Direct substitution.
5	Recognize the derivative: $\partial_t P = -\partial_y[a\,\Psi^\dagger Q\Psi]$.	Q and a have no y dependence.
6	Therefore the current is $J = a\,\Psi^\dagger Q\Psi$.	Compare with $\partial_t P + \partial_y J = 0$.
7	For the two-sector state, orthogonality gives $P = (1-p)\mu(y) + p\mu(y-G)$.	Cross terms vanish because $\langle\Phi_0\|\Phi_1\rangle = 0$.
8	Also $\Psi^\dagger Q\Psi = p\mu(y-G)$, so $J = a\,p\,\mu(y-G) = \dot{G}\,p\,\mu(y-G)$.	Q annihilates the unmarked sector and is identity on the marked sector.
9	The guidance law is $\dot{Y} = J/P$ wherever $P > 0$.	General current-over-density guidance equation.

[ ] Reproduce the continuity equation from the wave equation without using the result above.
[ ] Check that internal spin/logical indices are summed in $\Psi^\dagger\Psi$.
[ ] Confirm that P is normalized for every t.
[ ] Identify whether P can vanish on the chosen support and how measure-zero separatrices are handled.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.7. Conservation of cumulative Born probability

> **Critical verification gate**
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Define $C(y,t) = \int_{-\infty}^{y} P(z,t)dz$. The actual pointer coordinate is Y(t).

Step	Statement / calculation	Reason or check
1	Differentiate C with respect to time: $\partial_t C = \int_{-\infty}^{y} \partial_t P\, dz$.	Differentiate under the integral.
2	Use $\partial_t P = -\partial_z J$.	Continuity equation.
3	Integrate: $\partial_t C = -J(y,t) + J(-\infty,t)$.	Fundamental theorem of calculus.

Step	Statement / calculation	Reason or check
4	For a localized packet, $J(-\infty,t) = 0$, so $\partial_t C = -J$.	Boundary condition.
5	Also $\partial_y C = P$.	Differentiate the cumulative distribution with respect to its upper limit.
6	Along $Y(t)$, $dC/dt = \partial_t C + \dot{Y}\partial_y C$.	Chain rule.
7	Insert $\dot{Y} = J/P$: $dC/dt = -J + (J/P)P = 0$.	Exact cancellation.
8	Thus $u = C(Y(t),t)$ is constant along the trajectory.	This is the cumulative Born probability coordinate used in the manuscript.

[ ] Verify the boundary term for the specific packet family.

[ ] Confirm that “cumulative Born probability” is defined the first time it appears.

[ ] Explain in your own words why this is equivalent to non-crossing/order preservation in one dimension.

[ ] Check that no analogous simple scalar ordering is being assumed in higher-dimensional pointer space.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.8. The marked-state basin and deterministic ideal selection

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Step	Statement / calculation	Reason or check
1	At the final time, assume the unmarked and marked pointer packets are disjoint.	This is an explicit ideal measurement condition.
2	Assume the unmarked packet lies to the left and has total Born weight $1-p$.	Follows from its coefficient $\sqrt{(1-p)}$.
3	Assume the marked packet lies to the right and has total Born weight p.	Follows from its coefficient $\sqrt{p}$.
4	Every point in the left packet has final cumulative probability $C \leq 1-p$.	Integrating from $-\infty$ cannot include marked-packet weight.
5	Every interior point in the right packet has $C > 1-p$.	The cumulative distribution has already accumulated the full left-packet weight.
6	Because $u = C(Y(t),t)$ is conserved, the initial basin of the marked packet is $u > 1-p$.	Trace the final partition back to the initial time.
7	The boundary $u = 1-p$ is a measure-zero separatrix under a continuous pointer distribution.	Its treatment does not affect ensemble probabilities.

Step	Statement / calculation	Reason or check
8	A nonequilibrium ensemble supported wholly inside u > 1−p reaches the marked packet with probability one.	This is the ideal “unit probability” claim.

[ ] Check final packet separation quantitatively for the chosen packet and displacement.

[ ] Verify the inequality direction from the sign of G.

[ ] Check the no-instance case p = 0: there is no marked packet and Q acts as zero on the state.

[ ] State explicitly that “unit probability” is an ideal result conditional on exact preparation and good fault histories.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.9. Conditioned shell, normalization, and exact resource cost

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Use p* = 2^{-n} as the promised yes-instance design value. Define the same pointer ensemble whether the unknown instance is yes or no.

$$A = \{ y : 1 - p^*/2 < u(y) < 1 - p^*/4 \}$$

Step	Statement / calculation	Reason or check
1	Under the equilibrium measure μ, the cumulative variable u = C(y,0) is uniform on [0,1] for a continuous distribution.	Probability-integral transform.
2	The interval length is (1−p*/4) − (1−p*/2) = p*/4.	Direct subtraction.
3	Therefore μ(A) = p*/4.	Uniformity of u.
4	Every u in A satisfies u > 1−p*/2 > 1−p*.	Because p*/2 < p*.
5	Thus A lies strictly inside the marked-state basin of a unique yes instance.	Use the basin u > 1−p*.
6	Define ρ_A(y) = μ(y)1_A(y)/μ(A) = (4/p*)μ(y)1_A(y).	Condition μ on A.
7	Check normalization: ∫ρ_A dy = (4/p*)μ(A) = 1.	Substitute μ(A) = p*/4.
8	On A, ρ_A/μ = 4/p*; outside A, the ratio is zero.	Piecewise definition.

Step	Statement / calculation	Reason or check
9	Compute R = ∫ρ_A ln(ρ_A/μ)dy = ln(4/p*)∫ρ_A dy.	The likelihood ratio is constant on the support.
10	Therefore R = ln(4/p*) = n ln 2 + ln 4.	Substitute p* = 2^{-n}.
11	In bits, R/ln 2 = n + 2.	This is often the easiest interpretation.
12	The deterministic-selection lower bound is ln(1/p*) = n ln 2, so the construction is within exactly 2 bits of that lower bound.	Apply Module 5.3 with r = 1.

[ ] Verify directly that the shell is defined only from p*, not from the unknown value p_f.

[ ] Verify that the same ρ_{A*} is normalized and prepared on both promised cases before the verifier result is known.

[ ] Check the probability-integral transform assumptions: continuity and no atoms.

[ ] Confirm that the resource cost is reported consistently in nats or bits.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.10. Exponential-tail geometry with a smooth compact-support truncation

The exact Laplace calculation below isolates the geometry. To match the revised manuscript, place the shell inside an interval on which a compactly supported smooth packet has the same exponential survival law, and locate the smooth cutoffs a fixed distance outside the shell and final translated packet.

$$\mu(y) = (1/(2L)) \exp(-|y|/L), \quad L > 0$$

Step	Statement / calculation	Reason or check
1	For $y \geq 0$, the right-tail probability is $S(y) = \int_{y}^{\infty} \mu(z)dz = (1/2)e^{-y/L}$.	Integrate the exponential tail.
2	Since $u = 1-S(y)$, the shell condition is $p^*/4 < S(y) < p^*/2$.	Rewrite $1-p^*/2 < u < 1-p^*/4$.
3	At $S = p^*/2$: $(1/2)e^{-y_1/L} = p^*/2$, so $y_1 = L \ln(1/p^*)$.	Solve for the lower edge.
4	At $S = p^*/4$: $(1/2)e^{-y_2/L} = p^*/4$, so $y_2 = L \ln(2/p^*)$.	Solve for the upper edge.
5	The shell width is $y_2 - y_1 = L \ln 2$.	Independent of n.
6	For $p^* = 2^{-n}$, $y_1 = nL \ln 2$ and $y_2 = (n+1)L \ln 2$.	The shell moves linearly far into the tail.

Step	Statement / calculation	Reason or check
7	Thus location is O(n) and width is O(1).	This verifies the geometric claim for this packet family.

Worked check for n = 10 and L = 1:

- $p^* = 1/1024$.
- Shell equilibrium weight = $p^*/4 = 1/4096$.
- Lower edge $y_1 = 10 \ln 2 \approx 6.9315$.
- Upper edge $y_2 = 11 \ln 2 \approx 7.6246$.
- Width $y_2 - y_1 = \ln 2 \approx 0.6931$.
- Resource cost $R = \ln 4096 = 12 \ln 2 \approx 8.3178$ nats = 12 bits.

[ ] Check that the chosen wavefunction g satisfies $|g|^2 = \mu$ and is square-normalizable.

[ ] Construct or cite a smooth compactly supported density that agrees with the exponential-tail formula throughout the shell and its translated image; check normalization after adding the end caps.

[ ] Check that bounded interaction strength gives O(n) displacement time/segments and that these segments are included in L(n); retain the example as an illustration, not an experimental claim.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.11. Five-qubit perfect code and the complete syndrome decomposition

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

The revised manuscript avoids an abstract orthogonalization step by choosing the nondegenerate five-qubit [[5,1,3]] perfect code and the identity plus fifteen one-qubit Pauli errors.

Step	Statement / calculation	Reason or check
1	The physical Hilbert space of five qubits has dimension 2^5 = 32.	Five two-level systems.
2	The code space C encodes one qubit, so dim C = 2.	The code parameters are [[5,1,3]].
3	Use E_0 = I and the fifteen single-qubit Paulis X_j, Y_j, Z_j.	There are 5×3 = 15 nonidentity one-qubit Paulis.
4	For a nondegenerate distance-three code, P_C E_a†E_b P_C = δ_ab P_C for this set.	Distinct correctable Pauli errors have orthogonal syndromes.
5	Define V_a = E_aP_C. Then V_a†V_b = δ_abP_C.	Multiply the previous equation by the code projector.
6	Each range C_a = V_a C has dimension 2.	V_a is an isometry on C.
7	The sixteen ranges C_a are mutually orthogonal.	Their cross inner products vanish.

Step	Statement / calculation	Reason or check
8	Σ_a dim C_a = 16×2 = 32.	Dimension count.
9	Therefore K = ⊕_{a=0}^{15} C_a equals the full five-qubit Hilbert space.	An orthogonal 32-dimensional subspace of a 32-dimensional space is the whole space.

[ ] Read Laflamme et al. (1996) and independently verify the code parameters [[5,1,3]].

[ ] Verify nondegeneracy and the orthogonality relation for the identity plus fifteen one-qubit Paulis.

[ ] Verify the dimension count and that K is the entire physical Hilbert space for the perfect code.

[ ] Confirm E_0 = I and distinguish this exact five-qubit construction from the general nonperfect-code extension.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.12. Construction and properties of the error-transparent marked projector

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Let Q_L = |1_L⟩⟨1_L| act on C, let V_a = E_aP_C, and define the physical operator

$$\tilde{Q}_L = \Sigma_{a=0}^{15} V_a Q_L V_a^\dagger$$

Step	Statement / calculation	Reason or check
1	Each term V_aQ_LV_a† acts only on the orthogonal sector C_a.	V_a maps C isometrically into C_a.
2	(V_aQ_LV_a†)† = V_aQ_LV_a†.	Q_L is Hermitian.
3	Q̃_L† = Q̃_L.	Sum of Hermitian operators on orthogonal supports.
4	Q̃_LV_b = Σ_a V_aQ_LV_a†V_b = V_bQ_L.	Use V_a†V_b = δ_abP_C and Q_LP_C = Q_L.
5	$\tilde{Q}_L^2 = \Sigma_a V_a Q_L^2 V_a^\dagger = \tilde{Q}_L$.	Cross terms vanish and Q_L is a projector.
6	Therefore Q̃_L is a bounded self-adjoint projector with norm 1.	Nonzero orthogonal projector.
7	For the five-qubit perfect code, K is the full Hilbert space, so the formula is already global.	Module 5.11.
8	For a nonperfect fixed code, define Q̃_L = 0 on K⊥.	This supplies an explicit Hermitian extension.

Step	Statement / calculation	Reason or check
9	H_ET(t) = a(t)Q̃_Lp_y is self-adjoint on the natural domain of p_y when a(t) is real.	A bounded system operator tensor a self-adjoint momentum operator has the standard product-domain realization.
10	The body-ness is constant only because the base code has fixed size.	This is an asymptotic statement, not a two-body hardware construction.

[ ] Verify Hermiticity and idempotence explicitly.

[ ] Verify that the perfect-code formula acts on the full 32-dimensional space and that the general-code zero extension is explicit.

[ ] Confirm the Hamiltonian H_ET = aQ̃_Lp_y is self-adjoint on the intended domain.

[ ] Check that constant body-ness follows only because the base code size is fixed; it does not establish an easy two-body implementation.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.13. Single-fault error-transparent identity

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Let $U(t_2,t_1)=\exp\{-i[G(t_2)-G(t_1)]\tilde{Q}_{Lp_y}\}$. Suppose a one-qubit Pauli-basis fault E_a occurs at time t while the state was previously in the code sector.

Step	Statement / calculation	Reason or check
1	Before the fault, U(t,0)P_C = exp[−iG(t)Q_Lp_y]P_C.	On the code sector Q̃_L acts as Q_L.
2	The intertwining relation is Q̃_L E_aP_C = E_aQ_LP_C.	Because V_a = E_aP_C and Q̃_LV_a = V_aQ_L.
3	By induction, Q̃_L^mE_aP_C = E_aQ_L^mP_C for all m ≥ 0.	Repeated intertwining.
4	Functional calculus gives U(T,t)E_aP_C = E_a exp{−i[G(T)−G(t)]Q_Lp_y}P_C.	Expand the exponential; p_y commutes with the system operators.
5	Compose with the pre-fault evolution.	Insert U(t,0)P_C from Step 1.
6	U(T,t)E_aU(t,0)P_C = E_a exp{−i[G(T) −G(t)]Q_Lp_y}exp{−iG(t)Q_Lp_y}P_C.	Direct substitution.
7	Combine commuting logical exponentials to obtain E_a exp[−iG(T)Q_Lp_y]P_C.	The integrated couplings add.

Step	Statement / calculation	Reason or check
8	Every one-qubit operator is a linear combination of I,X,Y,Z on that qubit, so the identity extends linearly until syndrome resolution.	The Schrödinger evolution and the operator identity are linear.

Scope adopted in the revised manuscript

This identity proves every one-qubit fault history that begins in the code sector. The paper separately proves syndrome-only sequences. It does not call an arbitrary multi-fault history transparent; all histories outside the proved class are included in q_0.

[ ] Derive the identity using both a power series and direct action on logical eigenstates.

[ ] Check the time-dependent-a(t) version by splitting the integrated coupling before and after t.

[ ] Confirm that the base transparent set includes no fault and every one-qubit fault covered by the identity, and that no broader claim is made without proof.

[ ] Verify that faults in the pointer, syndrome apparatus, or coupling Hamiltonian are included in the gadget model or explicitly excluded.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____ _

5.14. Syndrome-only sequences and logical opacity

Critical verification gate

Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

This module verifies both the sequential transparency statement and the claim that the corresponding syndrome record does not reveal the logical marked/unmarked value.

Step	Statement / calculation	Reason or check
1	Define W by $WV_a\|\psi_L\rangle = \|\psi_L\rangle\otimes\|a\rangle$ on K.	The orthogonal syndrome sectors give a logical⊗syndrome factorization.
2	Then $W\tilde{Q}_L W^\dagger = Q_L\otimes I_{syn}$.	$\tilde{Q}_L$ applies the same logical projector in every syndrome sector.
3	If S preserves K and $WSW^\dagger = I_L\otimes s$, then $[S,\tilde{Q}_L]=0$ on K.	S acts only on the syndrome factor.
4	Any sequence of such S operators at arbitrary times can be commuted through the pointer propagators.	Each commutes with $Q_L\otimes I_{syn}$ and with p_y.
5	Hence a syndrome-only sequence produces exactly the ideal total pointer translation.	All intermediate faults can be moved to the end.

Step	Statement / calculation	Reason or check
6	For a correctable system–environment interaction, Knill–Laflamme makes the reduced syndrome/environment state independent of $\|\psi_L\rangle$.	The matrix elements reduce to c_ab, independent of the logical state.
7	Syndrome extraction may record the error label without recording the logical marked/unmarked value.	This is the logical-opacity condition needed before the nonequilibrium pointer acts.
8	Any history that leaves K, acts nontrivially on the logical factor, or fails the syndrome-only form is classified as nontransparent.	The theorem charges unproved cases to the failure event rather than assuming them harmless.

[ ] Construct W explicitly from the sixteen five-qubit syndrome sectors and verify $W\tilde{Q}_L W^\dagger = Q_L \otimes I_{syn}$.

[ ] Verify both the commutation argument for syndrome-only sequences and the Knill–Laflamme environment-independence calculation.

[ ] Read Beale et al., especially Theorem 2, and avoid claiming more than their local-noise assumptions establish.

[ ] Explain why ordinary dephasing that directly distinguishes logical 0 from logical 1 would violate the opacity assumption and destroy the intended advantage.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.15. Transported likelihood-ratio bound for bad events

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

The revised theorem specifies the full initial configuration q and conditions only the pointer coordinate: $\rho_0(q) = (4/p^*)1_{A^*}(y(q))\mu_0(q)$.

Step	Statement / calculation	Reason or check
1	Let q include computer, syndrome apparatus, modeled environment, and pointer; let $\mu_0(q)$ be the full equilibrium density.	The event bound must apply to the complete pilot-wave configuration.
2	Define $\rho_0(q) = (4/p^*)1_{A^*}(y(q))\mu_0(q)$.	Only the pointer marginal is conditioned.
3	Normalization: $\int \rho_0 dq = (4/p^*)\Pr_{\mu_0}(A^*) = 1$.	The equilibrium shell probability is $p^*/4$.
4	The initial likelihood ratio obeys $0 \le \rho_0/\mu_0 \le 4/p^*$.	It is $4/p^*$ on the shell and zero outside.
5	Equivariance transports the ratio unchanged along the full flow, preserving the essential supremum.	Module 5.1 applied to the joint system.

Step	Statement / calculation	Reason or check
6	For any later measurable event B, Pr_ρ(B) = ∫_B(ρ_t/μ_t)dμ_t.	Radon–Nikodym form of the nonequilibrium probability.
7	Bound the integrand to get Pr_ρ(B) ≤ (4/p*)Pr_μ(B).	Use the essential-supremum bound.
8	Since p*=2^{-n}, the maximum probability enhancement is 2^{n+2}.	The fault-tolerance suppression must overcome this factor.

[ ] Write the exact full joint initial density and verify its ratio bound.

[ ] Check that the system/environment are not silently assigned additional nonequilibrium.

[ ] Define every bad-history set B as an actual measurable configuration-space event, not only a formal fault-path label.

[ ] Verify assumption (A4): macroscopic syndrome records are disjoint enough to identify their Born weights with equilibrium event probabilities.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.16. Concatenated-code recursion and double-exponential suppression

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

Assume the stochastic, syndrome-resolved recursion $q_{j+1} \le Aq_j^2$ and define $\theta = Aq_0 < 1$.

Step	Statement / calculation	Reason or check
1	For $j = 0$, $q_0 = (1/A)\theta$.	Definition of θ.
2	At level 1, $q_1 \le Aq_0^2 = (1/A)\theta^2$.	Substitute $q_0 = \theta/A$.
3	Induction hypothesis: $q_j \le (1/A)\theta^{2^j}$.	Assume the claimed formula at level j.
4	Then $q_{j+1} \le Aq_j^2 \le A(1/A^2)\theta^{2\cdot 2^j}$.	Apply the recursion and square the induction bound.
5	Simplify: $q_{j+1} \le (1/A)\theta^{2^{j+1}}$.	Because $2\cdot 2^j = 2^{j+1}$.
6	Therefore $q_k \le (1/A)\theta^{2^k}$ for all k.	Mathematical induction.
7	Since $0 < \theta < 1$, θ^{2^k} decays doubly exponentially in k.	The exponent itself grows exponentially.

[ ] Verify AGP 2006, Eqs. (4)-(6), and note the independent stochastic-fault assumptions behind this simple probability recursion.

[ ] Do not apply this exact q recursion to an arbitrary coherent bath without an appropriate norm-amplitude theorem and a revised Bohmian-event argument.

[ ] Identify what q_0 means for the actual base interface gadget and how A is bounded.

[ ] Confirm $\theta = Aq_0$ is dimensionless and strictly below 1.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.17. Solving for the concatenation level k

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

For the complete L(n)-location circuit, require the nonequilibrium failure probability to be at most δ. It is sufficient that q_k ≤ δp*/[4L(n)].

Step	Statement / calculation	Reason or check
1	Use q_k ≤ A^{-1}θ^{2^k} and require A^{-1} θ^{2^k} ≤ δ2^{-n}/[4L(n)].	Combine the recursion with p*=2^{-n} and the full-circuit target.
2	Multiply by A: θ^{2^k} ≤ [Aδ/(4L(n))]2^{-n}.	A>0.
3	Take logs: 2^k ln θ ≤ ln[Aδ/(4L(n))] − n ln 2.	Natural logarithm is monotone.
4	Because ln θ<0, division reverses the inequality.	Critical sign check.
5	2^k ≥ [n ln 2 + ln(4L(n)/(Aδ))]/[−ln θ].	Algebraic rearrangement.
6	If L(n) is polynomial, ln L(n)=O(ln n), so the numerator is O(n).	The linear n ln 2 term dominates.
7	Therefore k=O(log n).	Take base-two logarithms.
8	If each level multiplies one-location cost by M, total cost L(n)M^k is polynomial.	M^{O(log n)}=n^{O(1)}.

Illustrative arithmetic check

- Choose A = 100, $q_0 = 0.001$, so θ = 0.1.
- Choose n = 100, δ = 0.01, and for illustration $L(n)=n^3=10^6$.
- Numerator: $100 \ln 2 + \ln[4\cdot10^6/(100\cdot0.01)] = 69.3147 + \ln(4\cdot10^6) \approx 84.5165$.
- Denominator: $-\ln 0.1 \approx 2.3026$.
- Required 2^k ≥ 36.70, so k = 6 is sufficient because $2^6 = 64$.
- This example is illustrative only; the actual A and q_0 must come from the specified gadget.

[ ] Redo the algebra by hand, paying special attention to the inequality reversal.

[ ] Check the calculation for the actual polynomial L(n), or leave the constants explicitly illustrative as in the manuscript.

[ ] Verify that the overhead exponent remains polynomial for the actual concatenated code.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL

Evidence / initials / date	_____

5.18. Complete-circuit bad event and good-history exactness

> **Critical verification gate**
>
> Do not sign the final amber declaration until this module passes or the manuscript is revised to remove the claim.

> **Revised manuscript status**
>
> The theorem now includes all L(n) logical locations and states as an assumption that a history with no incorrect or nontransparent logical location implements the ideal logical computation and transparent readout. Verify that this assumption matches the chosen fault-tolerant architecture.

Let B_k be the event that at least one of the L(n) logical locations is incorrect or nontransparent at concatenation level k.

Step	Statement / calculation	Reason or check
1	A1 includes state preparation, verifier, output-code transfer, syndrome operations, and bounded-strength pointer segments in L(n).	This closes the earlier interface-only gap.
2	For each location, the equilibrium bad probability is at most q_k.	Definition in A3.
3	$\Pr_\mu(B_k) \le L(n) q_k$.	Union bound; no independence is needed here.
4	On B_k^c, A3 asserts that the logical computation is ideal and the readout history is transparent.	This is the exactness assumption required for a Bohmian-basin claim.
5	Unique witness: the pointer begins in $u>1-p^*$ and enters the marked packet.	Ideal pointer result.
6	No witness: $p_f=0$, so the marked packet is absent and the current is zero.	Separate no-instance analysis.
7	Therefore only B_k can cause a wrong promised-case output.	Good histories are exactly correct by assumption.
8	The transported likelihood ratio gives $\Pr_\rho(\text{failure}) \le 4L(n)q_k/p^*$.	Combine Modules 5.15 and the union bound.

[ ] Verify that every physical operation named in A1 is represented by at least one logical location in L(n).

[ ] Check that bounded-strength segmentation of an O(n) pointer displacement is included in L(n).

[ ] Verify or explicitly accept A3's exact good-history statement; ordinary diamond-norm closeness alone is not a substitute.

[ ] Confirm that pointer faults, correlated control faults, and all histories outside the proved transparent set are included in q_0.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

5.19. Complexity-theoretic consequence

The revised manuscript now states a randomized decision consequence and treats witness recovery separately. Verify each logical implication rather than relying on a slogan about NP search.

Step	Statement / calculation	Reason or check
1	Unsatisfiable input: every formula produced by the isolation reduction is unsatisfiable.	Required for soundness on zero instances.
2	Satisfiable input: with inverse-polynomial probability, a derived formula has exactly one satisfying assignment.	The isolation guarantee.
3	The promise solver may behave arbitrarily on derived formulas with more than one solution.	A promise-problem solver has no obligation off the promise.
4	Repeat the isolation experiment polynomially many times and amplify the bounded-error promise solver.	Raises the inverse-polynomial success probability to a constant or smaller error.
5	This yields a randomized polynomial-time SAT decision procedure in the enlarged subquantum model.	The conclusion is randomized and model-relative.
6	Recover a witness by SAT self-reduction: fix variables one at a time using repeated decision queries.	Decision-to-search is a separate standard argument.

[ ] Read the original Valiant-Vazirani paper, not only a secondary summary.

[ ] Record the precise promise problem: distinguish unsatisfiable instances from instances with exactly one satisfying assignment.

[ ] Record the type of randomized reduction and its success probability.

[ ] Check how many repetitions are needed to raise the isolation success probability to a constant.

[ ] Verify the variable-fixing SAT self-reduction and the polynomial number of decision calls needed to recover a witness.

[ ] Confirm that all language says randomized polynomial time in the enlarged subquantum model, not that ordinary BPP contains NP.

> **Revised manuscript formulation**
>
> SAT randomized-reduces to repeated zero-versus-one isolation instances. After amplification, the assumed subquantum primitive gives a randomized SAT decision procedure in the enlarged computational model; a satisfying assignment follows by standard self-reduction. No deterministic collapse is claimed.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

6. Primary-source and citation audit

For amber classification, references generated or surfaced during AI-assisted work must be independently checked. A DOI resolving is not enough: verify that the paper supports the exact claim, record the page/equation, and note any limitations.

S1. Springer Nature, "AI use in research practice" (current policy page)

https://www.springernature.com/gp/policies/editorial-policies/using-ai-in-research

Claims in the manuscript that this source is being used to support:

- Amber use may include AI contributions to analytical or interpretive work.
- Human validation, methodological description, transparency, and accountability are required.

Human verification:

[ ] Read the complete page on the date of submission.

[ ] Save a PDF or screenshot showing the traffic-light table.

[ ] Confirm that the final declaration describes both purpose and validation.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

S2. Springer Nature, "AI use in manuscript preparation"

https://www.springernature.com/gp/policies/editorial-policies/ai-manuscript-preparation

Claims in the manuscript that this source is being used to support:

- AI may assist structuring or shaping text under amber conditions.
- AI must not replace authorship or scholarly judgement.

Human verification:

[ ] Check the current wording immediately before submission.

[ ] Confirm that the abstract and declaration do not understate the amount of AI drafting.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

S3. A. Valentini, "De Broglie-Bohm Quantum Mechanics," arXiv:2409.01294 / Encyclopedia of Mathematical Physics

https://arxiv.org/abs/2409.01294

Claims in the manuscript that this source is being used to support:

- Pilot-wave continuity and guidance equations.
- Measurement interaction $H_I = a\,\omega\,p_y$ and translated packets.
- $df/dt = 0$ for $f = \rho/|\psi|^2$.
- The open question about polynomial versus exponential nonequilibrium-resource scaling.

Human verification:

[ ] Check PDF pages 2-7 and 12 (printed pages 2, 5, 7, 12).

[ ] Record Eqs. (2)-(5), (18)-(21), and the exact open-question passage.

[ ] Confirm the 2025 encyclopedia bibliographic metadata and DOI.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____ _

S4. A. Valentini, "Subquantum Information and Computation," Pramana 59, 269-277 (2002)

https://arxiv.org/abs/quant-ph/0203049

Claims in the manuscript that this source is being used to support:

- Nonequilibrium apparatus distributions and subquantum measurement.
- The proposal to use such measurement computationally.

Human verification:

[ ] Read Sections 4 and 7 in the PDF.

[ ] Check that the manuscript does not attribute the new tail-conditioned construction to Valentini.

[ ] Record the exact scope of Valentini's computational claim.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____ _

S5. A. Valentini, "Signal-locality, uncertainty, and the subquantum H-theorem. I," Phys. Lett. A 156, 5-11 (1991)

https://doi.org/10.1016/0375-9601(91)90116-P

Claims in the manuscript that this source is being used to support:

- Historical source for quantum nonequilibrium and the coarse-grained H-theorem.

Human verification:

[ ] Verify title, volume, pages, year, and DOI.

[ ] Check whether it is necessary for each sentence currently citing it, or whether the review is the direct source.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____ _

S6. R. Laflamme, C. Miquel, J. P. Paz, and W. H. Zurek, "Perfect Quantum Error Correcting Code," Phys. Rev. Lett. 77, 198–201 (1996)

https://doi.org/10.1103/PhysRevLett.77.198

Claims in the manuscript that this source is being used to support:

- The five-qubit [[5,1,3]] code is a perfect nondegenerate code correcting every one-qubit error.

- The identity and fifteen one-qubit Pauli errors generate sixteen mutually orthogonal two-dimensional syndrome sectors, exhausting the 32-dimensional physical Hilbert space.

Human verification:

[ ] Read the original code construction and verify the stated parameters [[5,1,3]].

[ ] Check that the code is nondegenerate for the chosen correctable Pauli set.

[ ] Verify the dimension count 16 × 2 = 32 and the claim that the syndrome sectors exhaust the five-qubit Hilbert space.

[ ] Confirm that the manuscript does not attribute the error-transparent pointer construction itself to this source.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

S7. E. Knill and R. Laflamme, "Theory of Quantum Error-Correcting Codes," Phys. Rev. A 55, 900-911 (1997)

https://arxiv.org/abs/quant-ph/9604034

Claims in the manuscript that this source is being used to support:

- Necessary and sufficient correctability conditions.
- Logical-state-independent syndrome/environment information.

Human verification:

[ ] Read Theorem 3.2 and Eqs. (19)-(20).

[ ] Read the tensor-product syndrome characterization in the later part of Section 3.

[ ] Verify the logical⊗syndrome factorization and the claim that correctable syndrome information is independent of the logical state.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

S8. O. Vy, X. Wang, and K. Jacobs, "Error-transparent evolution," New J. Phys. 15, 053002 (2013)

https://arxiv.org/abs/1210.8007

Claims in the manuscript that this source is being used to support:

- Error-transparent evolution replicates the same logical Hamiltonian in each error space.
- Higher-body interactions may be required.

Human verification:

[ ] Check Eq. (1) and the construction H = H_0 + Σ_i H_i.

[ ] Check the body-ness results and do not generalize them beyond their assumptions.

[ ] Confirm the author name is Os Vy.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL

Evidence / initials / date	_____

S9. P. Aliferis, D. Gottesman, and J. Preskill, "Quantum accuracy threshold for concatenated distance-3 codes," QIC 6, 97-165 (2006)

https://arxiv.org/abs/quant-ph/0504218

Claims in the manuscript that this source is being used to support:

- For independent stochastic faults, bad exRec probabilities obey a quadratic recursion and fall doubly exponentially below threshold.

Human verification:

[ ] Read printed page 10, Eqs. (4)-(6).

[ ] Record the exact stochastic and independence assumptions.

[ ] Check whether the paper's more general noise results use amplitudes/norms rather than probabilities and avoid conflating the two.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

S10. S. J. Beale et al., "Quantum Error Correction Decoheres Noise," Phys. Rev. Lett. 121, 190501 (2018)

https://arxiv.org/abs/1805.08802

Claims in the manuscript that this source is being used to support:

- Syndrome measurements suppress coherent off-diagonal logical noise for local noise in stabilizer codes.

Human verification:

[ ] Read Theorem 2 and its local-noise assumptions.

[ ] Verify the scaling O(r^{d/2}) and what is averaged over syndromes.

[ ] Do not use this as a blanket theorem for arbitrary coherent baths.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

S11. L. G. Valiant and V. V. Vazirani, "NP is as easy as detecting unique solutions," TCS 47, 85-93 (1986)

https://doi.org/10.1016/0304-3975(86)90135-0

Claims in the manuscript that this source is being used to support:

- Zero-versus-unique SAT and unique-solution search are hard under randomized reductions.

Human verification:

[ ] Read the original paper.

[ ] Record the exact reduction and promise behavior.

[ ] Check the decision-to-search wording used in the manuscript.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

7. Novelty and prior-art search

AI-assisted literature discovery is not a substitute for a systematic human search. Run and record the following queries in at least two scholarly databases, preferably Google Scholar plus arXiv/Web of Science/Scopus or MathSciNet where available.

[ ] "subquantum computation" resource scaling

[ ] "quantum nonequilibrium" computational resource

[ ] Bohmian nonequilibrium computation complexity

[ ] pilot-wave rare branch measurement nonequilibrium

[ ] Bohmian measurement basin cumulative probability

[ ] nonequilibrium relative entropy resource theory pilot wave

[ ] error-transparent logical measurement pointer

[ ] error-transparent Hamiltonian quantum measurement

[ ] fault-tolerant nonlinear quantum computation

[ ] Valentini computation NP nonequilibrium precision

[ ] rare-event reweighting Bohmian measurement

[ ] conditional Born distribution tail pointer hidden variables

Search log template

Date/database	Query	Closest result	Decision / reason

[ ] Identify the closest prior construction to the cumulative-probability-conditioned pointer.
[ ] Identify any prior paper using relative entropy as a pilot-wave nonequilibrium resource measure.
[ ] Identify prior fault-tolerant nonlinear or hidden-variable computation proposals.
[ ] State the novelty claim narrowly enough to survive the closest result found.
[ ] Retain dated search results as evidence.

Result	The revised lemma has the needed hypotheses. Verify the monotonicity proof and note that deterministic selection has lower bound ln(1/p*) while the construction uses ln(4/p*).
Evidence / initials / date	_____

8. Claim-strength and terminology audit

“Unit probability”	Retain only with the explicit design-shell, one-dimensional guidance, packet-separation, ideal no-instance, and good-history qualifications.
“Matching lower bound”	The revised lemma has the needed hypotheses. Verify the monotonicity proof and note that deterministic selection has lower bound ln(1/p*) while the construction uses ln(4/p*).
“Transparent history”	Use only for the one-fault histories and syndrome-only sequences explicitly proved. Every other history is nontransparent and contributes to q_0.
“Fault-tolerant search”	The revised theorem includes the complete L(n)-location circuit. Verify A1–A4 and do not extend the claim to unsyndromed coherent baths.
“Polynomial total overhead”	The proof includes L(n), code overhead M^k, and O(n) pointer segmentation. It excludes the physical cost of producing/maintaining nonequilibrium and realizing the fixed higher-body interface.
“Marked state”	Use this for the unique witness basis state; use “marked subspace” for the Q=1 sector and “marked-state pointer packet” for the translated outcome.
“Cumulative Born probability”	Define C(y,t) explicitly; avoid the unexplained phrase “Born quantile.”
“Resource”	State that R is an information-theoretic additive measure, not a complete laboratory cost.
“Answers Valentini’s question”	Prefer “answers a precise conditional version” because production, maintenance, and microscopic hardware realization remain open.
Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL
Evidence / initials / date	_____

9. Human contribution, validation, and AI disclosure

The purpose of this section is not to minimize the AI contribution. It is to document the human work that makes the use defensibly amber rather than substitutive.

Human contribution ledger

Contribution	Record
Research question	Author supplied Valentini's open scaling question as the target.
Conceptual direction	Author introduced and developed the branchwise nonlinear-gate idea that motivated the search for a digital amplifier.
Critical scrutiny	Author repeatedly challenged proposed constructions, including precision and fault-tolerance weaknesses.
Mathematical validation	To be completed: author independently re-derives and signs every module in this dossier.
Source validation	To be completed: author checks every primary source and citation claim.
Novelty assessment	To be completed: author performs and records the independent literature search.
Scope and assumptions	Author decides whether A1-A4, the base interface, and the resource definition are scientifically acceptable and how narrowly to state the theorem.
Final interpretation	Author decides what the result means and which conclusions are warranted.
Writing and voice	AI supplied extensive draft text; author revises the final paper into language and argument that the author fully endorses.
Accountability	Author is sole author and accepts responsibility for accuracy, integrity, originality, and corrections.

Suggested post-verification disclosure

> **Use only after every statement is true**
>
> Most exploratory derivations and much of the initial manuscript drafting were produced through extended interactive use of OpenAI ChatGPT (GPT-5.6 Sol). ChatGPT proposed intermediate constructions, including the cumulative-probability-conditioned pointer argument and the error-transparent fault-tolerance analysis. The author independently re-derived all mathematical results, checked all cited primary sources, performed an independent novelty search, corrected and narrowed claims where required, determined the final assumptions, interpretation, and conclusions, and takes full responsibility for the accuracy, integrity, and originality of the work. ChatGPT is not listed as an author.

[ ] The disclosure names the tool and model.

[ ] The disclosure states the purposes: exploration, analytical suggestions, and drafting.

[ ] The disclosure identifies substantive AI contributions rather than calling them mere language editing.

[ ] The disclosure states the independent validation steps actually completed.

[ ] The disclosure states that final judgement and accountability are human.

[ ] The abstract and end declaration are consistent with one another.

Result	[ ] PASS [ ] REVISION NEEDED [ ] FAIL

Evidence / initials / date	_____

10. Final amber-readiness sign-off

> **This is a human attestation, not an AI-generated conclusion**
>
> Sign only after completing the revised dossier and retaining the evidence. A repaired manuscript is not yet a verified manuscript. If any statement below is not true, revise the paper or do not claim amber readiness.

[ ] I can reproduce the central pointer and fault-tolerance derivations without consulting ChatGPT.

[ ] I have independently reproduced the revised rare-set proof, including the hypotheses, derivative sign, endpoint, and asymptotics.

[ ] I have checked every cited source and recorded exact supporting locations.

[ ] I have conducted and retained an independent novelty search.

[ ] I understand and accept every assumption, including the full-circuit L(n) model, the definition of transparent histories, and the limits of A1–A4.

[ ] I have decided the final theorem statement, scope, interpretation, and conclusion myself.

[ ] I have verified the five-qubit syndrome decomposition, the global definition of $\tilde{Q}_L$, the one-fault identity, and the syndrome-only sequential argument.

[ ] I have checked the union-bound factor L(n), the likelihood-ratio enhancement 4/p*, and the logarithmic concatenation-level calculation.

[ ] I have verified the randomized Valiant–Vazirani consequence and the separate SAT self-reduction used for witness recovery.

[ ] The AI disclosure is complete and does not minimize the model's contribution.

[ ] I accept sole responsibility for the submitted work.

Overall decision	[ ] READY FOR AMBER SUBMISSION [ ] NOT YET READY
Author name	Michael A. Stay
Signature	
Date	
Evidence archive location	

Appendix A. Compact equation dependency map

→ Continuity equations → f = ρ/μ is advected → fine-grained R is invariant.

→ Verifier state → p* = 2^{-n}, witness count w ∈ {0,1}, and actual marked weight p_f = wp*.

→ H_M = aQp_y → translated marked packet → density P and current J.

→ Continuity + guidance → cumulative Born probability u is conserved.

→ Final packet ordering → marked basin u > 1−p*.

→ Condition μ on shell A → deterministic ideal selection and R = ln(4/p*).

→ Data processing → lower bound Ω(ln(1/p*)) = Ω(n).

→ Five-qubit [[5,1,3]] code → sixteen orthogonal two-dimensional syndrome sectors that exhaust the 32-dimensional physical space.

→ Replicated projector $\tilde{Q}_L$ → global bounded projector, one-fault error-transparent translation, and syndrome-only sequential transparency.

→ Logical opacity + A3–A4 → histories with no incorrect or nontransparent logical location implement the ideal computation and basin.

→ Likelihood-ratio bound → nonequilibrium bad-event probability ≤ (4/p*) times equilibrium weight.

→ Concatenation recursion → $q_k \le (1/A)\theta$ raised to the power 2^k.

→ Union bound $\Pr_{\mu}(B_k) \le L(n)q_k$ and event bound $\Pr_{\rho}(B_k) \le (4/p^*)\Pr_{\mu}(B_k)$ → k = O(log n), with total cost $L(n)M^k$ = poly(n).

→ Randomized isolation + bounded-error promise solver → randomized SAT decision in the enlarged model; SAT self-reduction → witness recovery.

Appendix B. Minimal arithmetic-check code

Running code is not a substitute for understanding the derivation, but the following short script can catch transcription and sign errors after the algebra has been checked by hand.

```
import math

def binary_kl(r, p):
    term1 = 0.0 if r == 0 else r * math.log(r / p)
    term2 = 0.0 if r == 1 else (1-r) * math.log((1-r) / (1-p))
    return term1 + term2

def derivative_binary_kl(r, p):
    return math.log(r * (1-p) / (p * (1-r)))

def resource_cost(n):
    p_star = 2.0 ** (-n)
    return math.log(4.0 / p_star), n + 2.0

def required_level(n, A, q0, logical_locations, delta=0.01):
    theta = A * q0
    assert 0 < theta < 1
    rhs = (n*math.log(2) + math.log(4*logical_locations/(A*delta))) / (-math.log(theta))
    return math.ceil(math.log2(max(1.0, rhs)))

# Monotonicity to the right of p.
p, c = 2.0**-20, 0.1
assert derivative_binary_kl(c, p) > 0
assert binary_kl(0.2, p) >= binary_kl(c, p)

# Revised full-circuit example.
n = 100
L = n**3
print(resource_cost(n))
print(required_level(n=n, A=100, q0=0.001,
                     logical_locations=L, delta=0.01))  # 6
```

Appendix C. Revision crosswalk and remaining verification

Implemented: Lemma 1 now states $0<p<c\leq 1$ and $r\geq c$, proves monotonicity of $d(r||p)$ for $r>p$, and handles $c=1$ by continuity.

Implemented: $p^*=2^{-n}$ is the design value and $p_f=wp^*$ is the actual marked-state weight.

Implemented: the five-qubit perfect-code sectors exhaust the physical Hilbert space; the general fixed-code version is extended by zero on $K^{\perp}$.

Implemented: the paper proves every one-qubit fault identity, proves syndrome-only sequential transparency, defines those histories as transparent, and charges all others to q_0.

Implemented: A1 defines the complete $L(n)$-location circuit and the theorem uses $\Pr\rho(\text{failure})\leq 4L(n)q_k/p^*$.

Implemented: A3–A4 explicitly assume syndrome-resolved probability events and exclude a general unsyndromed coherent bath.

Implemented: the paper states a randomized SAT decision consequence and separately invokes standard SAT self-reduction for witness recovery.

Still required before submission: after completing this dossier, update the AI disclosure only with validation steps that were actually performed and retain the supporting evidence.

[ ] Independently reproduce every revised derivation rather than treating the manuscript repair as verification.

[ ] Verify the new five-qubit-code citation and every source location supporting the revised claims.

[ ] Complete the novelty search, including the cumulative-probability shell and error-transparent readout combination.

[ ] Decide whether the fixed-size higher-body interface and A3–A4 are scientifically acceptable assumptions for the submitted claim.

Appendix D. Source list

- Springer Nature, AI use in research practice. https://www.springernature.com/gp/policies/editorial-policies/using-ai-in-research
- Springer Nature, AI use in manuscript preparation. https://www.springernature.com/gp/policies/editorial-policies/ai-manuscript-preparation
- Springer Nature, editor AI assessment guidance. https://www.springernature.com/gp/policies/editorial-policies/ai-editor-guidance
- Valentini, De Broglie-Bohm Quantum Mechanics, arXiv:2409.01294. https://arxiv.org/abs/2409.01294
- Valentini, Subquantum Information and Computation, quant-ph/0203049. https://arxiv.org/abs/quant-ph/0203049
- Valentini, Signal-locality, uncertainty, and the subquantum H-theorem I. https://doi.org/10.1016/0375-9601(91)90116-P
- Laflamme, Miquel, Paz, and Zurek, Perfect Quantum Error Correcting Code. https://doi.org/10.1103/PhysRevLett.77.198
- Knill and Laflamme, A Theory of Quantum Error-Correcting Codes. https://arxiv.org/abs/quant-ph/9604034
- Vy, Wang, and Jacobs, Error-transparent evolution. https://arxiv.org/abs/1210.8007
- Aliferis, Gottesman, and Preskill, Quantum accuracy threshold. https://arxiv.org/abs/quant-ph/0504218
- Beale et al., Quantum Error Correction Decoheres Noise. https://arxiv.org/abs/1805.08802
- Valiant and Vazirani, NP is as easy as detecting unique solutions. https://doi.org/10.1016/0304-3975(86)90135-0